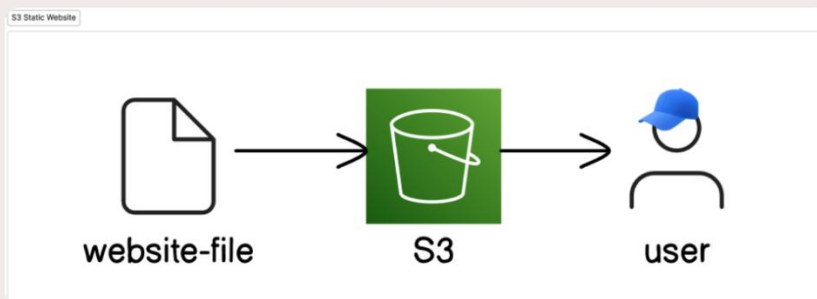

Host a Website on Amazon S3

- **KEY CONCEPTS: Amazon S3**

You can use Amazon S3 to store and retrieve any amount of data at anytime from anywhere on the web. In today's project, we'll be creating an S3 bucket to store pictures and files needed in your very own website. Amazon S3 (which stands for Amazon Simple Storage Service) to host a website.

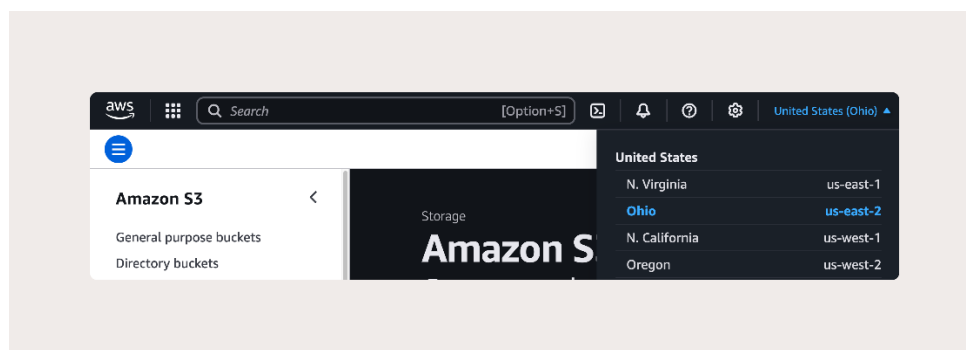
💡 **Websites** are run by pictures and files. You'll be hosting (which basically means make public on the Internet) a website on Amazon S3 by uploading website content into S3 and then making it publicly accessible.



Create a bucket in Amazon S3

Alright! Let's start by opening up an iconic AWS service called **Amazon S3** 📁. This is where you'll store the files that make up your website.

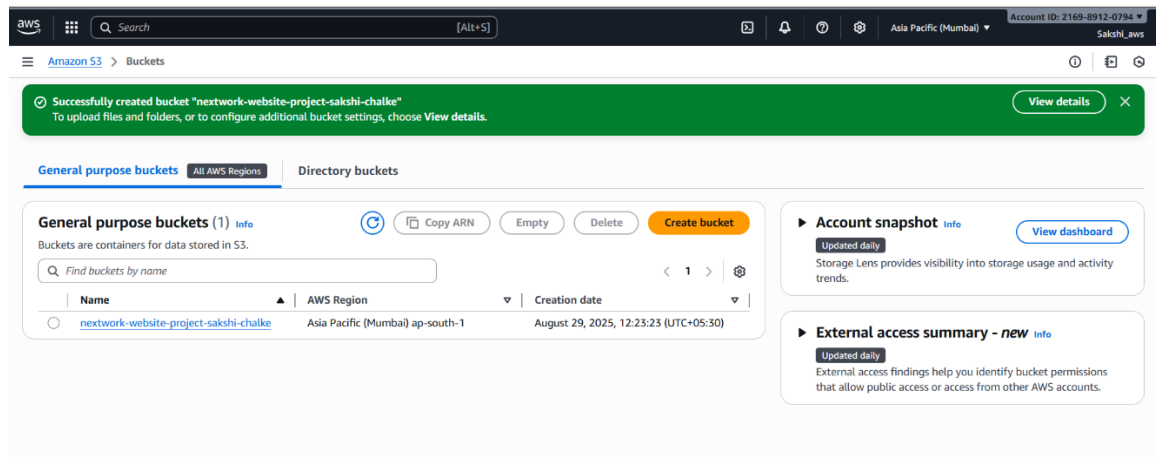
- Log in to the **AWS console**
- In the AWS Management Console, search for S3.
- Select the **AWS Region** closest to you. You can find this at the top right corner of your AWS Management console, right next to your name.



- Choose **Create bucket**.
- For **Bucket name**, enter : nextwork-website-project-enter your name
- For **Object Ownership**, choose **ACLs enabled**.

💡 Why do I have to enter my name in the bucket name?

An S3 bucket name is globally unique. After you create a bucket, **no other AWS account in the entire world** can use your bucket's name (unless you delete the bucket).



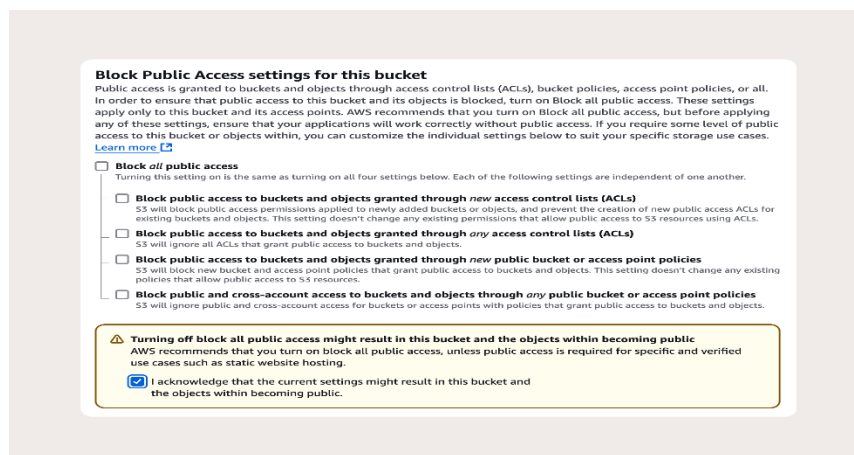
What are ACLs (Access Control Lists)?

An **ACL** = a set of rules that decides who can get access to a resource.

Enabling ACLs in this S3 setup lets you control who can access and do things with the objects (i.e. website files) you upload into your bucket.

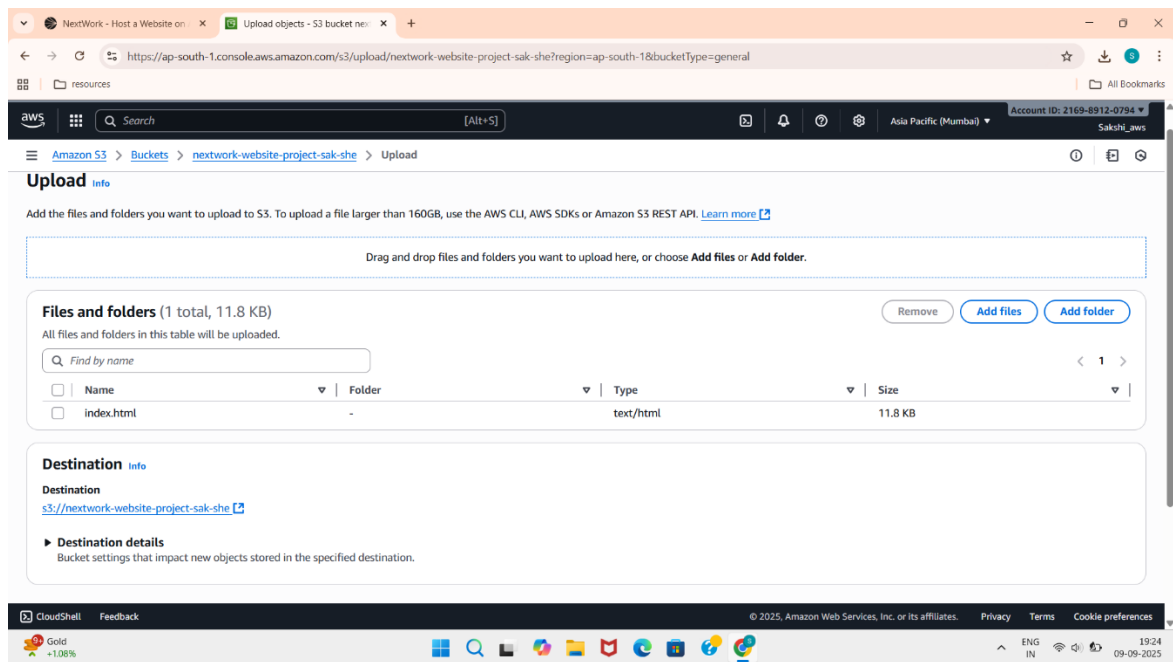
With ACLs, different AWS accounts can own and control different files in your bucket.

- Choose **Bucket owner preferred**.
- For **Block Public Access settings for this bucket**, clear the check box for **Block all public access**.
- Check the box that says **“I acknowledge that the current settings might result in this bucket and the objects within becoming public.”**
- For **Bucket Versioning**, choose **Enable**.
- Choose **Create bucket**.



Upload website content to your bucket

- Download an HTML file that sets up your website
 - .Download a zip file of images for your website.
 - Upload both files into your S3 bucket.
 - In the **Buckets** section, choose the name of your new bucket.
 - Download these files to your computer so we can add them into your bucket (right click on each link, and select **Save link as...**):
 - [index.html](#)
-
- Return to the Amazon S3 console with your bucket page open. Choose the **Objects** tab.
 - Choose **Upload**.
 - Choose **Add files**.
 - Choose **index.html**.
 - Choose **Upload** in your AWS console.



Configure a static website on Amazon S3

S3 bucket? Created ☒

Website files? Uploaded ☒

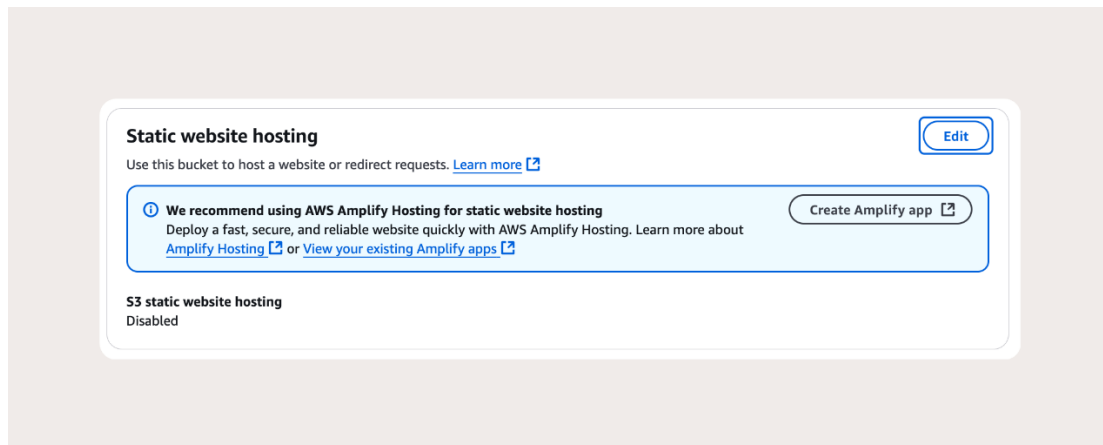
Next up, let's make your website available on the internet by setting up **static website hosting**!

TASKS

- Configure your S3 bucket for static
- website hosting Visit your public website link.

In your EC2 console, choose **Launch instance**.

- Make sure you're back in your bucket's page. If you're not sure, choose **General purpose buckets** on the left hand side navigation bar, and then choose the bucket you created for this project.
- Choose the **Properties** tab.
- Scroll alllllllll the way down to the **Static website hosting** panel.
- Choose **Edit**.
- Choose **Save changes**.
- In the **Static website hosting** panel, click on the URL under **Bucket website endpoint**.



Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

- ☐ Disable
☒ Enable

Hosting type

- ☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document

Specify the home or default page of the website.

index.html

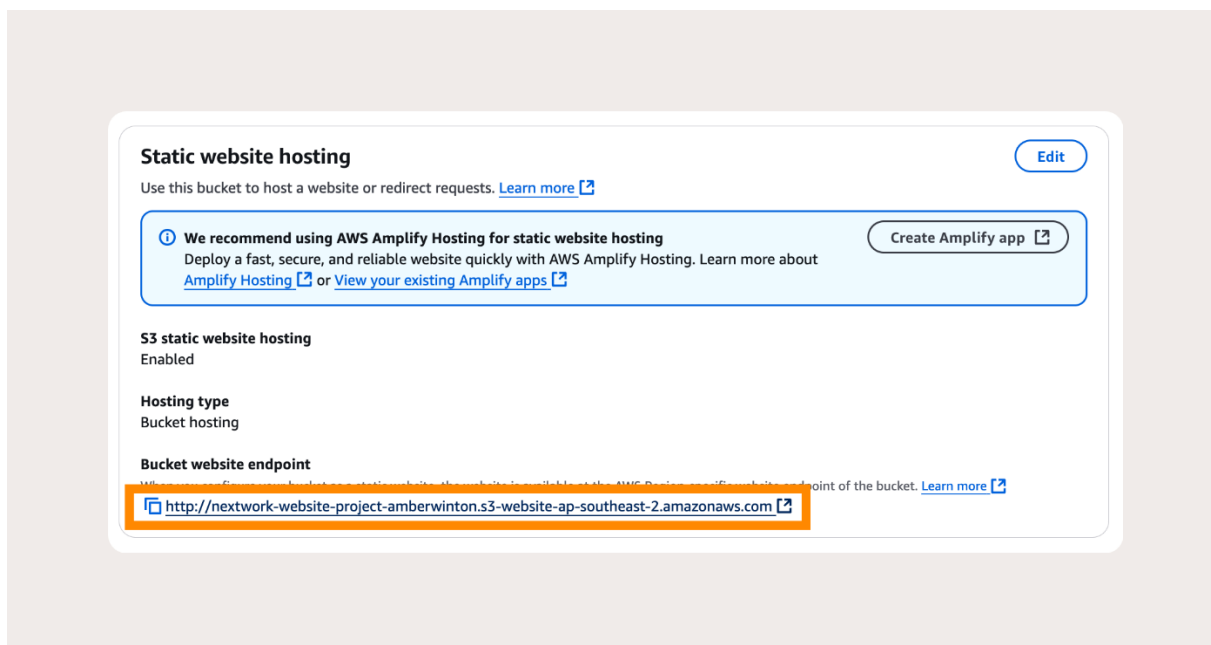
Error document - optional

This is returned when an error occurs.

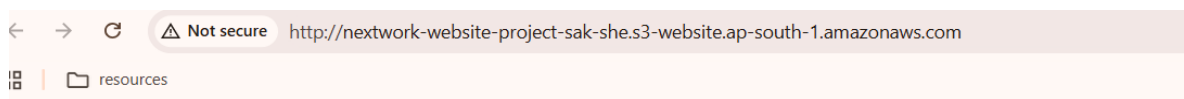
error.html

💡 What's a bucket website endpoint?

A bucket website endpoint is just like a regular website URL. It lets people visit your S3 bucket's files as a website.



An error! 🙄



103 Forbidden

- Code: AccessDenied
- Message: Access Denied
- RequestId: YN9KXS2XBA3QVTYG
- HostId: p4lhtq3JCKyg8gUUrqtTj79+e5yjL4HV3Y9eRSZFfxVP+EWnXEwbzoAlzH/5ysCDnbKouiqDsuQ=

💡 Why did I get this error?

Objects (in this case, the HTML and images files you uploaded) are private by default. This default setting helps keep your account's data secure.

The error message you're seeing is telling you that your static website is being hosted by S3, but the actual HTML/image files you've uploaded are still private. It's kind of like having a bucket on display, so everyone can see the bucket - but the contents are covered up, preventing anyone from seeing what's inside.

To solve this error, we need to set the permission of the objects to public - this is why we enabled ACLs in Task 1!

Make objects in your S3 bucket public

- Keep the error message tab open and switch back to the Amazon S3 console tab.
- Do you still remember how to view your S3 bucket's objects? Try finding your bucket's **Objects** page and making your objects public using ACLs.

